

Research Paper

Effect of micro encapsulated phase change material on the anti-dry-out ability of pulsating heat pipes

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ABSTRACT

Anti-dry-out ability is one of the key performance indexes of the Pulsating Heat Pipe (PHP). In order to improve the anti-dry-out ability, the Micro Encapsulated Phase Change Material (MEPCM) suspension is added to the closed PHP. The Computational Fluid Dynamics (CFD) technique is used to establish the numerical model and analyze the anti-dry-out ability of the PHPs with water and with MEPCM suspension as the working fluids respectively. By changing the phase change temperature range of the MEPCM suspension, the effect of the MEPCM suspension with different phase change temperatures on the anti-dry-out ability, start-up performance and heat transfer performance of the PHP is further studied. The results show that the MEPCM suspension within a certain phase change temperature range can effectively improve the anti-dry-out ability of the PHP. The appropriate phase change temperature range should be selected around the evaporation section temperature when the PHP transforms from the circulating flow state to the dry-out state. After the MEPCM suspension is used as the working fluid, the start-up time and thermal resistance of the PHP are also increased, but the changes are not significant.

1. Introduction

With the development of semiconductor technology, various electronic equipment have entered people's lives extensively, and the trend of miniaturization and integration greatly increases the power consumption and heat generation per unit area of these electronic equipment. So it is urgent to strengthen the research and applications of heat transfer devices of high thermal conductivity. In order to meet the work and production requirements of higher heat load, smaller design space and lower cost, the development of passive cooling system with higher performance has become a focus of current research [1,2]. As a new heat transfer device, the PHP has a series of advantages such as high heat transfer efficiency, good adaptability, simple structure and low cost. So since it was invented in the 1990s [3], the PHP has been paid close attention to by researchers due to its unique working principle and excellent heat transfer efficiency [4]. And it has been applied in a variety of fields, such as heat dissipation of electronic equipment [5–9], heat collection of solar energy [10,11], recovery of waste heat [12], thermal management of power units [13–16] and heat dissipation of on-orbit applications [17,18] and so on.

A typical PHP structure consists of an evaporation section, an adiabatic section and a condensation section. With the temperature increase of the evaporation section, the motion states of the working fluid in the PHP can be divided into non-oscillation state, intermittent oscillation state, continuous oscillation state of one-way circulation, circulating flow state and dry-out state [19]. When partial dry-out phenomenon occurs in the PHP, the heat transfer efficiency of the PHP will reach its maximum value, called the heat transfer limit. When the working fluid in the pipe is completely dried out, the thermal resistance will rise sharply, resulting in a significant decrease in heat transfer efficiency and function failure of the PHP [20]. Therefore, improving the anti-dry-out ability (or heat transfer limit) of the PHP will help to expand its operation temperature range and enhance its environmental adaptability.

There are many design and use parameters which can affect the anti-dry-out ability of the PHP, such as length, cross section shape, Filling Rate (FR), inclination angle, input power and type of the working fluid, etc. Wang et al. [21] studied the effect of the condensation section length on the anti-dry-out ability of the PHP. The results showed that the PHP with a condensation section length of

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Nomenclature

k	thermal conductivity (W/m·K)
t	time (s)
$\vec{v}$	the velocity vector (m/s)
S_m	mass source term due to phase change (kg/m ³ ·s)
S_k	latent heat source term due to phase change (W/m ³ ·s)
T	temperature (K)
P	pressure (Pa)
$\vec{F}$	interface volumetric force (N/m ³)
C	surface contact angle
C_p	specific heat (J/kg·K)
c	volume fraction of MEPCM particles in suspension
c	mass fraction of MEPCM particles in suspension
E	internal energy (J/kg)
h_{LH}	latent heat (J/kg)
Q	input power (W)
R	filling rate
R_{th}	thermal resistance (K/W)

Greek letters

ρ	density (kg/m ³)
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μ	dynamic viscosity (Pa·s)
α	volume fraction
σ	surface tension coefficient (N·m)

Subscripts

0	initial value
a	apparent (no phase change)
b	MEPCM suspension
c	condenser
e	evaporator
l	liquid
v	vapor
w	water
MPL	lower limit
MPH	upper limit
mp	MEPCM particle
mix	mixture
sat	saturation

100 mm began drying at an input power of 120 W, however when the condensation section length was 50 mm, it began drying at an input power of 160 W. Tseng et al. [22] changed the cross section of the adiabatic section of the PHP to an elliptical shape and conducted experiments. The results showed that the pipe with an elliptical shape could generate additional unbalanced pressure distribution, making the working fluid easier to circulate, and the anti-dry-out ability was better. Saha et al. [23] found that the ideal FR of the PHP was between 30 and 70% by experiments. When the filling rate was lower or higher, the dry-out phenomenon would occur in advance. Shi et al. [24] studied the influence of the inclination angle on the heat transfer limit of the PHP. It was found that with the decrease of the inclination angle, the circulation effect of the working fluid in the pipe became weaker, which resulted in an increase in the thermal resistance of the PHP and a lower heat transfer limit. Wang et al. [25] found by experiments that the dry-out phenomenon of non-azeotropic working fluid appeared later than the working fluid of single medium. And when the difference between the boiling points of the two kinds of working fluid was larger, the effect was more significant. Hu et al. [26] found in experiments that the surface tension of self-wetting fluid increased with temperature. When used as the working fluid for the PHP, it could spontaneously replenish the liquid film evaporated by heat, and thereby improve the heat transfer efficiency, and delay the occurrence of the dry-out phenomenon. Karthikeyan et al. [27] analyzed the anti-dry-out ability of

deionized water and nanoparticle fluid as the working fluid of PHPs by experiments. It was found that when the input power reached 180 W, the deionized water was completely dried out and the temperature rose sharply, but the nanoparticle fluid was not completely dried out until the input power reached 240 W. It was because the nanoparticle helped the heat to dissipate rapidly by increasing the thermal conductivity and nucleate boiling point of the working fluid. And the heat transfer limit was increased by 33.3%.

However, the research on the anti-dry-out ability/heat transfer limit of the PHPs is mainly based on experimental methods in the past. With the development of CFD technique, especially the multiphase flow theory and the Volume Of Fluid (VOF) method, as well as the improvement of computer processing capacity, the CFD technique has been used more and more in the operation process and performance analysis research of the PHPs. For example, Wang et al. [28] studied the start-up performance of a single-loop PHP by establishing a CFD model. Pouryoussefi et al. [29] used CFD technique to study the effect of input power and liquid FR on the thermal resistance of the PHP. E Jiaqiang et al. [30] studied the effect of pipe diameter on the heat transfer efficiency of a single-loop PHP. However, as the PHP needs much longer time to reach the dry-out state, the amount of computation and time cost will also be greater. By now, the research on the performance analysis of the PHP by CFD technique mainly focuses on the start-up performance and heat transfer performance.

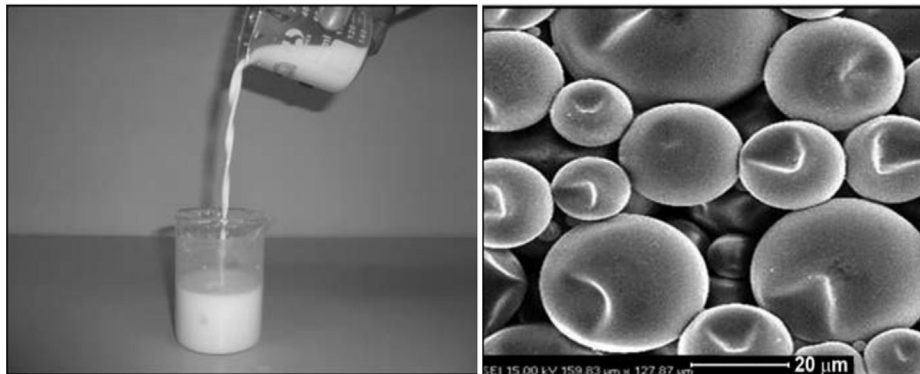


Fig. 1. Appearance and scanning electron microscopy photograph of MEPCM suspension [31].

The MEPCM suspension is a novel latent heat fluid with high heat storage density (Fig. 1). It is made by adding a certain proportion of MEPCM to the base fluid to form the suspension. The heat-absorption and heat-release by the phase change process of the Phase Change Material (PCM) can significantly increase the equivalent specific heat of the suspension in a certain temperature range, while the liquidity of the base fluid can be still maintained [32,33]. By changing the type of PCM and the mass fraction of the MEPCM, the phase change temperature range and equivalent specific heat of the MEPCM suspension can be adjusted conveniently. As shown in Fig. 1, the base fluid of the MEPCM suspension is composed of water and acetone. The MEPCM is composed of a melamine resin shell and a n-hexadecane core, and the phase change temperature range of the MEPCM is (324–334) K. When the mass fraction of the microcapsules is 20%, the heat storage density of the MEPCM suspension can be 30 J/g.

2. Mathematical model

2.1. Governing equations for the PHP

The unsteady flow of vapor phase and liquid phase in the PHP can be modeled by the VOF method. This method divides the fluid into multiple analysis domains and defines the volume fraction of the vapor phase and liquid phase on each domain. The density, thermal conductivity and dynamic viscosity of each domain are obtained by multiplying the density, thermal conductivity and dynamic viscosity of the vapor phase and the liquid phase by the volume fraction (Eq. (1)). Substituting the above parameters into the governing equations and solving them, we can get the function of the volume fraction of the vapor phase and liquid phase with time, and track the movement of the vapor and liquid interface and the change of working parameters according to this function [34].

$$\rho = \alpha_l \rho_l + \alpha_v \rho_v, k = \alpha_l k_l + \alpha_v k_v, \mu = \alpha_l \mu_l + \alpha_v \mu_v \quad (1)$$

where ρ is the density, k is the thermal conductivity, μ is the viscosity, and α is the volume fraction. The subscript l represents the liquid phase, and v represents the vapor phase.

The vapor phase and the liquid phase satisfy the continuous Eqs. ((2) and (3)), respectively.

$$\frac{\partial \alpha_l}{\partial t} + \nabla \cdot (\alpha_l \vec{v}) = \frac{S_{m,l}}{\rho_l} \quad (2)$$

$$\frac{\partial \alpha_v}{\partial t} + \nabla \cdot (\alpha_v \vec{v}) = \frac{S_{m,v}}{\rho_v} \quad (3)$$

where t is time, $\vec{v}$ is the velocity vector, S_m is the source term of mass transfer between liquid and vapor phase, which can be calculated by Eqs. (4) and (5) [35,36].

$$S_{m,l} = \begin{cases} -0.1 \alpha_l \rho_l \left| \frac{T_{mix} - T_{sat}}{T_{sat}} \right| & T \geq T_{sat} \\ 0.1 \alpha_v \rho_v \left| \frac{T_{mix} - T_{sat}}{T_{sat}} \right| & T < T_{sat} \end{cases} \quad (4)$$

$$S_{m,v} = \begin{cases} 0.1 \alpha_l \rho_l \left| \frac{T_{mix} - T_{sat}}{T_{sat}} \right| & T \geq T_{sat} \\ -0.1 \alpha_v \rho_v \left| \frac{T_{mix} - T_{sat}}{T_{sat}} \right| & T < T_{sat} \end{cases} \quad (5)$$

where T_{mix} represents the temperature of mixed phase, T_{sat} represents the saturation temperature.

The momentum equation is shown in Eq. (6).

$$\frac{\partial}{\partial t}(\rho \vec{v}) + \nabla(\rho \vec{v} \vec{v}) = -\nabla p + \nabla[\mu(\nabla \vec{v}^T)] + \rho \vec{g} + \vec{F} \quad (6)$$

where p is the pressure. A Continuous Surface Tension (CST) model is used to treat the surface tension $\vec{F}$ between vapor and liquid, as shown in Eq. (7) [37].

$$\vec{F} = \sigma \frac{\alpha_l \rho_l C_v \nabla \alpha_v + \alpha_v \rho_v C_l \nabla \alpha_v}{0.5(\rho_l + \rho_v)} \quad (7)$$

where C is the surface contact angle, and σ is the surface tension coefficient.

The energy equation is shown in Eq. (8).

$$\frac{\partial}{\partial t}(\rho E) + \nabla[\vec{v}(\rho E + p)] = \nabla(k \nabla T) + S_h \quad (8)$$

where E is the internal energy defined by the saturation temperature T_{sat} and the specific heat capacity at constant pressure C_p , as shown in Eq. (9).

$$E = \frac{\alpha_l \rho_l C_{p,l}(T - T_{sat}) + \alpha_v \rho_v C_{p,v}(T - T_{sat})}{\alpha_l \rho_l + \alpha_v \rho_v} \quad (9)$$

where S_k is the energy source term caused by the phase change, as shown in Eq. (10).

$$S_k = h_{LH} S_{m,l} = h_{LH} S_{m,v} \quad (10)$$

where h_{LH} is the latent heat of the working fluid.

In this research, the PHP thermal resistance is defined as Eq. (11).

$$R_{th} = (T_e - T_c)/Q \quad (11)$$

where T_e and T_c are the average temperature of the evaporation section and the condensation section, respectively, and Q is the input power.

2.2. Governing equations for the MEPCM

MEPCM suspension is a kind of solid-liquid multiphase fluid, it is formed by mixing MEPCM and single-phase heat transfer fluid such as water. Compared with ordinary single-phase heat transfer fluid, MEPCM suspension retains the function of heat storage and heat transfer of the PCM, and unifies the function of heat storage material and heat transfer fluid. Its thermophysical properties can be calculated by the thermophysical properties of the MEPCM particles and the single-phase fluid.

The overall density of the MEPCM suspension depends on the density of the MEPCM particles, the density of the single-phase fluid, and the volume fraction of the MEPCM particles in the suspension. In this research, MEPCM particles are considered uniformly dispersed in water.

$$\rho_b = \rho_w(1 - c^*) + \rho_{mp} c^* \quad (12)$$

where ρ_b is the overall density of the MEPCM suspension (liquid phase), ρ_w is the density of water, ρ_{mp} is the particle density of the MEPCM, and c^* is the volume fraction of MEPCM particles in the suspension. As the water in the PHP is evaporated, the MEPCM particles are gradually concentrated in the suspension, so c^* varies with the volume fraction of the liquid phase in the PHP, as shown in Eq. (13). Since the working fluid in the PHP circulates continuously in the pipe, here it is assumed that all the working fluid is uniformly evaporated and condensed.

$$c^* = c_0^* R / \alpha_l \quad (13)$$

where c_0^* is the initial volume fraction of the MEPCM particles in the suspension, and R is the FR.

The latent heat caused by the phase change process is characterized by the difficulty of temperature change in a certain temperature range, and its effect is equivalent to the change of the specific heat capacity of the MEPCM suspension. Roy et al. [38] proposed the equivalent specific heat model of MEPCM suspension. In the model, the MEPCM suspension is considered as uniform fluid, and the latent heat is equivalent to the change of the specific heat capacity at constant pressure, and the equivalent specific heat is assumed as constant.

$$C_{p,b} = \begin{cases} C_{p,a} & T < T_{MPL} \\ C_{p,a} + \frac{ch_{LH,mp}}{T_{MPL} - T_{MPH}} & T_{MPL} < T < T_{MPH} \\ C_{p,a} & T > T_{MPH} \end{cases} \quad (14)$$

where $C_{p,b}$ is the overall specific heat capacity at constant pressure of the MEPCM suspension (liquid phase). c is the mass fraction of the MEPCM particles in the suspension, which also varies with the volume fraction of the liquid phase in the PHP, as shown in Eq. (15). $h_{LH,mp}$ is the latent heat of MEPCM particles. T_{MPL} and T_{MPH} are the lower and upper limits of the phase change temperature range, respectively. $C_{p,a}$ is the apparent specific heat, that is, the specific heat exhibited by the MEPCM suspension out of the phase change temperature range, as shown in Eq. (16) [38].

$$c = c_0 \frac{R\rho_{b0}}{\alpha_l \rho_b} \quad (15)$$

where c_0 is the mass fraction of the initial MEPCM particles in the suspension, ρ_{b0} is the overall density of the initial MEPCM suspension.

$$C_{p,a} = C_{p,w}(1 - c) + C_{p,mp}c \quad (16)$$

where $C_{p,w}$ is the specific heat of water, $C_{p,mp}$ is the specific heat of a single MEPCM particle.

The overall thermal conductivity k_b of the MEPCM suspension is given by the Maxwell equation [39].

$$\frac{k_b}{k_w} = \frac{2 + \frac{k_{mp}}{k_w} + 2c^* \left(\frac{k_{mp}}{k_w} - 1 \right)}{2 + \frac{k_{mp}}{k_w} - c^* \left(\frac{k_{mp}}{k_w} - 1 \right)} \quad (17)$$

where k_w is the thermal conductivity of water, k_{mp} is the thermal conductivity of a single MEPCM particle.

The overall dynamic viscosity μ_b of the MEPCM suspension is given by the Vand viscosity equation [39]

$$\frac{\mu_b}{\mu_w} = (1 - c^* - 1.16c^{*2})^{-2.5} \quad (18)$$

where μ_w is the dynamic viscosity of water.

In this research, the MEPCM prepared in literature [31] is used, and the microcapsule material is composed of n-hexadecane and melamine resin, and water is used as the base fluid. Substituting the relevant material parameters into the above equations, the initial values of the relevant physical properties of the MEPCM suspension can be obtained, as shown in Table 1. Since the condensation section of the PHP is assumed to be the cooling section at constant temperature, the heat released during the phase change of the MEPCM from liquid to solid does not affect the temperature of the condensation section, so it is not necessary to set the phase change temperature range when the PCM changes from liquid phase to solid phase.

By substituting the data in Table 1 into Eq. (13), the initial value $C_{p,b0}$ of the equivalent specific heat capacity $C_{p,b}$ can be obtained, as shown in Fig. 2. As can be seen from the figure, the equivalent specific heat of the MEPCM suspension with a mass fraction of 20% is greatly increased compared to the water in the temperature range of (324–334) K, which is caused by the phase change process of the PCM in the MEPCM. However, in other temperature ranges, the equivalent specific heat of the MEPCM suspension with a mass fraction of 20% is slightly lower than that of the water, it is mainly because the apparent specific

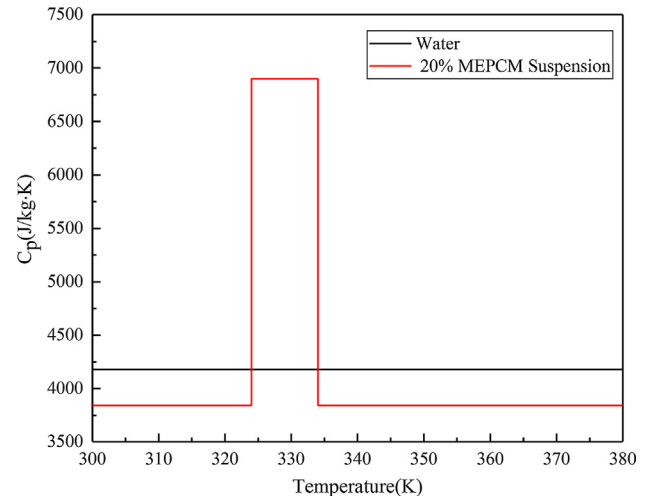


Fig. 2. Initial equivalent specific heat capacity $C_{p,b0}$ of the MEPCM suspension.

heat of the MEPCM is less than that of the water.

3. Physical model and model verification

3.1. Physical model and numerical verification

A two-dimensional model of the PHP is established in ANSYS as shown in Fig. 3. The PHP is placed vertically, and the upper end is the condensation section, the middle part is the adiabatic section, and the lower end is the evaporation section. Water is used as the working fluid first. The liquid phase is defined as the primary phase and the vapor phase is the secondary in the VOF model, and the FR is 60%. T1, T2, and T3 are temperature monitor points of the evaporation section, respectively. In order to simplify the present model and reduce the computation workload, the following assumptions are made for the object: (1) The wall thickness is neglected. (2) The vapor phase is all water vapor. (3) The vapor phase is the ideal vapor. (4) The liquid phase density and specific heat capacity of the water do not change with the temperature. The simulation process is divided into two steps. Firstly, in order to simulate the initial vapor-liquid distribution in the PHP, the initial boundary conditions of constant temperature at 303.15 K, and sub-atmospheric pressure in the whole computational domain are assumed. The contact angle of the liquid phase and the pipe wall is 30°, and the vapor-liquid surface tension coefficient σ is 0.075 N/m. Under the effect of wall adhesion and surface tension, the internal working fluid will gradually form an intermittent distribution state of the vapor plugs and liquid columns. Then, the evaporation section is changed to constant heat flux density boundary condition, and the heat flux density is calculated by dividing the input power by the surface area of the evaporation section. The adiabatic section is set to adiabatic boundary condition, and the temperature of the condensation section maintains 303.15 K. A structured grid is used, encrypted near the wall, and the number of grids is determined to be 48,025 through independence verification.

In the numerical simulation, the pressure-velocity coupling adopts the PISO algorithm which introduces the proximity correction and the

Table 1
Initial values of thermophysical properties of MEPCM and its suspension.

	density /(kg/m ³)	apparent specific heat /(J/kg.K)	thermal conductivity /(W/m.K)	viscosity /(kg/m.s)	latent heat /(kJ/kg)	phase change temperature range from solid phase to liquid phase /(K)
water	997	4180	0.604	1.0×10^{-3}	2357.5	–
MEPCM	900	2500	0.117	–	152.8	324–334
20% suspension	975	3844	0.469	2.24×10^{-3}	–	–

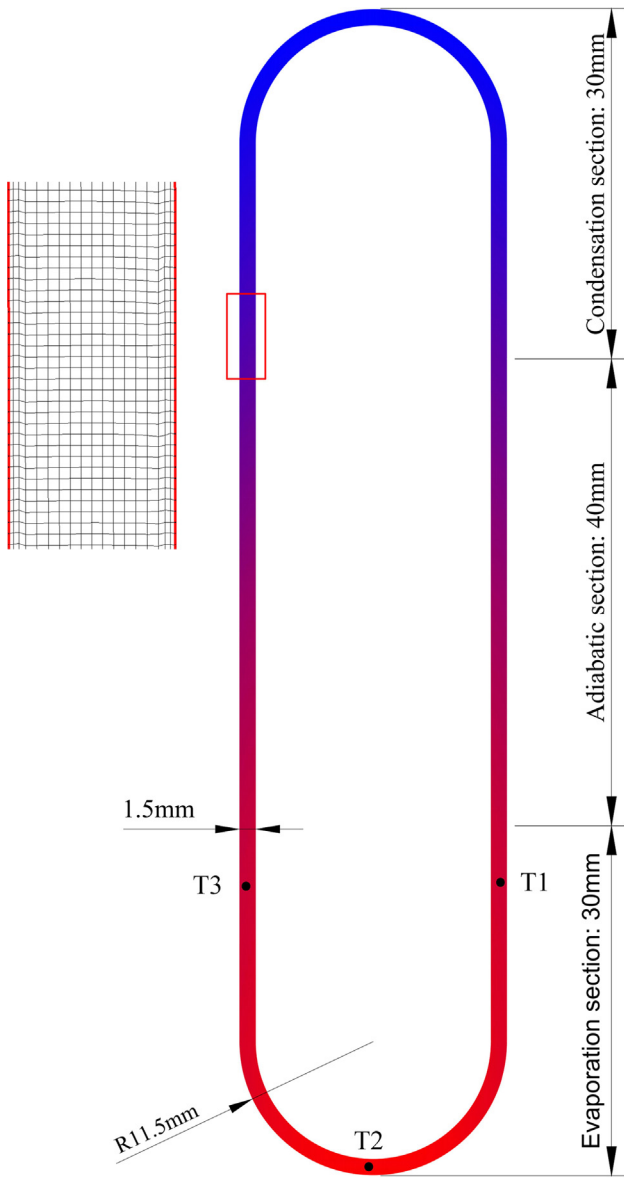


Fig. 3. Schematic diagram of the physical model of a single-loop PHP.

skewness correction, and the momentum and energy equations adopt the second-order upwind style discrete method. In order to better track the vapor-liquid surface, the volume fraction is discretized by the Geo-Reconstruct discretization method. The Eqs. (4 and 5) are respectively compiled by the user-defined function (UDF) and loaded into the quality source item of the vapor phase and the liquid phase. The Eq. (10) is compiled by the UDF and loaded into the energy source term of the mixed phase. In order to verify the accuracy of the model, the experimental study of the PHP in the literature [23] is carried out by numerical simulation. The PHP shape parameters and experimental conditions are consistent with those in the literature. The comparison between the calculation results and the experimental results is shown in Fig. 4. The average error between the calculated thermal resistance of the PHP and the experimentally measured value is 6.08%. It can be considered that the modeling method is reasonable and reliable.

3.2. Temperature changes and state transform of the working fluid

The state transform of the working fluid will make the surface temperature of the PHP show different change laws, so that the wall temperature becomes an important basis for identifying the motion

state of the working fluid in the PHP [40]. Fig. 5 shows the temperature changes of T1, T2, and T3 at the input power of 60 W. It can be seen that the temperature at T1, T2 and T3 show the same characteristics, that is, as the time increases, the temperature of the evaporation section increases continuously until the “peak” is reached, then the temperature begins to decrease and a gradually decaying oscillation state is formed. Since the internal working fluid flows clockwise, the temperature of each measuring point is $T1 < T2 < T3$, and the phase angle is $T1 > T2 > T3$, which is consistent with the experimentally observed law in the literature [41]. The average temperature T_e of the evaporation section at this power can be obtained by averaging the temperatures at T1, T2, and T3. Since the change pattern of the average temperature T_e tends to be steady after the “peak”, it can be generally divided into the start-up state and the steady operation state after the “peak”, as shown in Fig. 5. As the condensation section is at constant temperature of 303.15 K and the evaporation section is at constant input power of 60 W, according to Eq. (11), the variation of PHP thermal resistance with time is consistent with the average temperature of the evaporation section.

The CFD technique can simulate the generation and growth of bubbles, the combination and reduction of vapor plugs, and the state transform process of the working fluid. The start-up state and operation state of the PHP at the input power of 60 W are shown in Fig. 6(a). In this figure, “t = 12 s” represents the initial vapor-liquid distribution. It can be observed that the vapor plugs and the liquid columns are intermittently distributed in the pipe. The contact angle is formed between the liquid column and the pipe wall, and the liquid film is formed between the vapor plug and the pipe wall. As the heating progresses, the working fluid is heated in the evaporation section, and tiny bubbles are generated (C), the bubble merges with the bubble (D), the bubble merges with the vapor plug (E), the vapor plug grows (A1-A2-A3), and the working fluid is pushed clockwise to the condensation section. In the condensation section, the vapor plug is cooled, causing its liquefaction and shrinkage (B1-B2-B3). Eventually, because the input power is too high, there's not enough time for the heat to dissipate. The temperature of the PHP keeps rising, and dry-out phenomenon begins. This is consistent with the experimental result obtained in the literature [42].

4. Results and discussion

4.1. Effect of MEPCM on start-up performance and heat transfer performance

Based on the model settings above, the volume fraction of the liquid

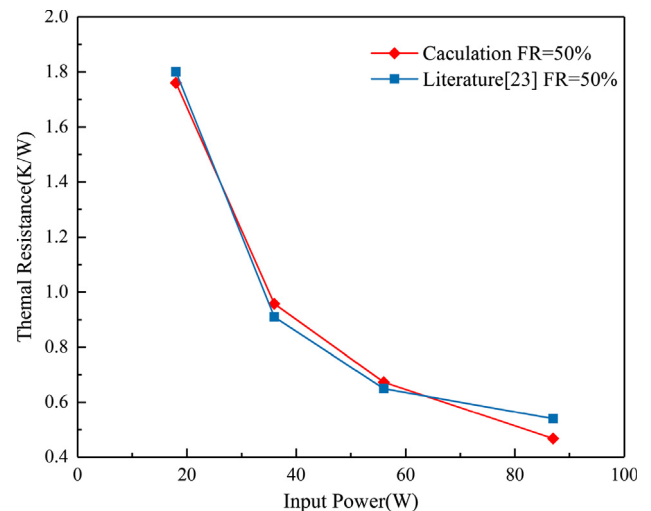


Fig. 4. Comparison of calculation results with experimental results.

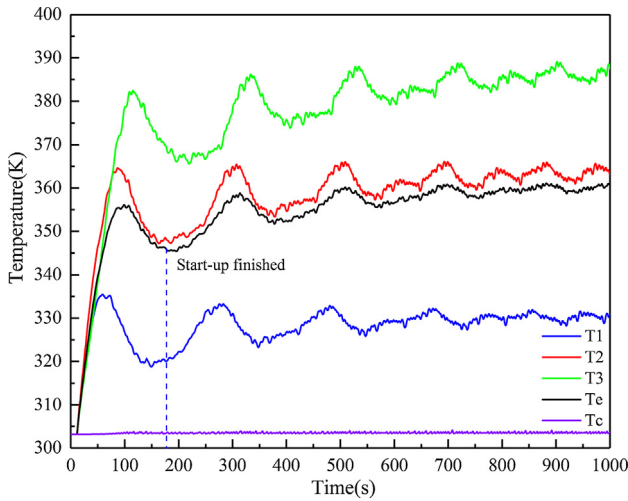


Fig. 5. The temperature changes of the evaporation section and the condensation section when the input power is 60 W.

phase in each element is integrated by the ADJUST macro in the ANSYS Fluent User Defined Function (UDF) to obtain the volume fraction of the liquid phase α_l in the pipe. Then bring it into Eqs. (13 and 15) to obtain the volume fraction c^* and mass fraction c of the MEPCM, and write the UDF in combination with Eqs. ((12), (14), (17) and (18)), and then load it into the density ρ_b , specific heat capacity $C_{p,b}$, thermal conductivity k_b , and viscosity μ_b of the liquid phase. The MEPCM suspension is heated at 60 W to obtain the start-up time and thermal resistance of the PHPs (Fig. 7).

The start-up time and thermal resistance are important indexes for evaluating the start-up performance and heat transfer performance of PHPs. The shorter the start-up time, the better the start-up performance. The smaller the thermal resistance, the better the heat transfer performance. The start-up time and the thermal resistance of the circulating flow state at 1000 s of the above described PHPs vary with different kinds of the working fluid, as shown in Fig. 7. After adding MEPCM in the PHPs, the start-up time of all the PHPs is prolonged. The start-up time of the PHPs with MEPCM suspensions as the working fluid is increased by 2.2–18.9% compared with that of the PHP with water. The start-up time of the PHP with the MEPCM suspension of the phase change temperature range of (324–334) K is the longest, this is mainly because the heat-absorption by the phase change process of the MEPCM during the start-up process delays the temperature raise of the evaporation section.

After the MEPCM suspension is used as the working fluid, the thermal resistances of all the PHPs are slightly increased. The observed minimum and maximum thermal resistances are 0.99 K/W and 1.06 K/W, respectively, compared with 0.95 K/W of the PHP with water. This is maybe because the density and the apparent specific heat of the MEPCM suspension are both less than those of the water. The MEPCM suspensions with higher phase change temperature range, including (364–374 K, 374–384 K, 404–414 K), have not undergone phase change at 1000 s yet, so the average temperatures of the evaporation section are higher, and the thermal resistances are greater. As for the MEPCM suspensions with lower phase change temperature ranges, including (324–334 K, 354–364 K), the average temperatures of the evaporation sections are still higher than that of the PHP with water, resulting in greater thermal resistance than that of the PHP with water, but lower thermal resistance than those of the PHPs with the MEPCM suspensions with higher phase change temperature ranges above. Although the temperature rising rate can be reduced to some extent by the heat absorption effect of the PCM, it is still not enough to offset the effect of the reduction of the overall density and apparent specific heat capacity of the working fluid in this circumstance, due to the narrower phase

change temperature range and lower mass fraction of the MEPCM.

4.2. Effect of MEPCM on anti-dry-out ability

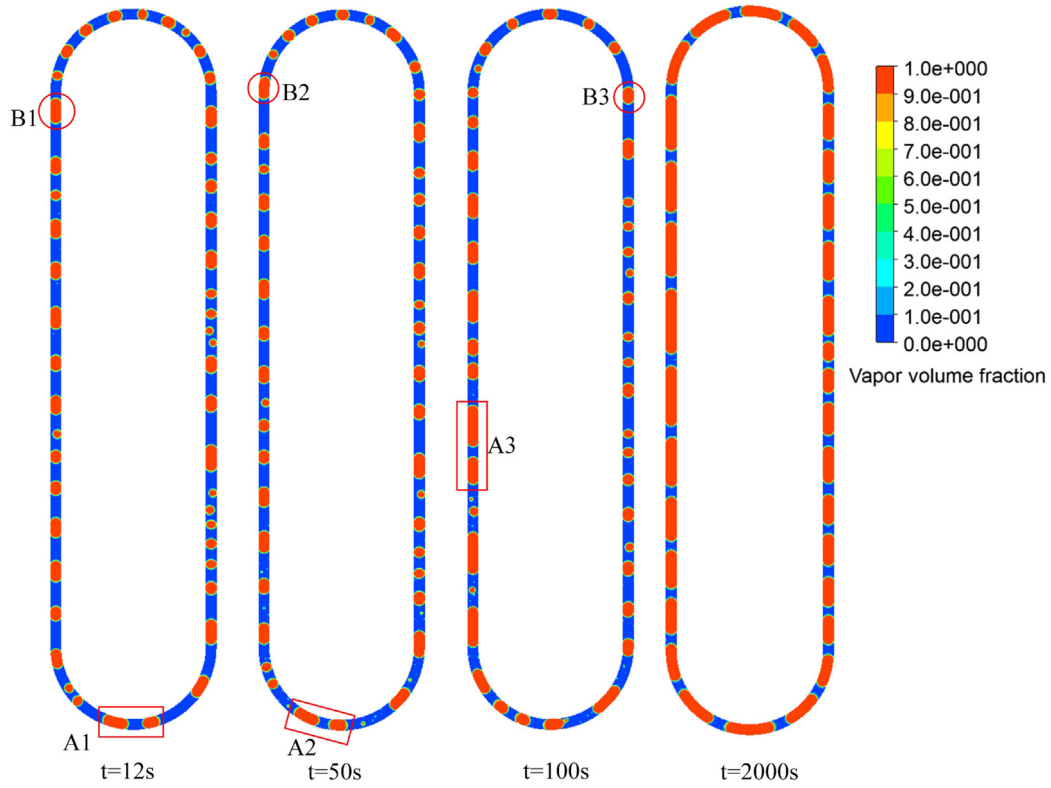
The MEPCM suspension is heated at 60 W to obtain the average temperature of the evaporation section (Fig. 8). As can be seen from the figure, at the same input power, the temperature of the evaporation section rises faster than that of the PHP with water. This is because the overall density ρ_b and the apparent heat capacity $C_{p,a}$ of the working fluid are less than those of the water after the addition of MEPCM. When the average temperature reaches 324 K, the MEPCM starts to absorb heat through phase change, and the temperature rising rate decreases. At about 326 K, the temperature is already lower than the water. When the average temperature reaches 334 K, the phase change of the MEPCM is completed, and the temperature rising rate starts to increase again, resulting in an increase of the “peak” amplitude, and the temperature in the steady operation state is slightly higher than that of the PHP with water.

With the increase of heating time, the PHPs with water and with MEPCM suspension as the working fluid both begin to transform from the circulating flow state to the dry-out state at around 1400 s. Since the variation of the PHP thermal resistance is consistent with the average temperature of the evaporation section, the PHPs with water and with MEPCM suspension as the working fluid also reach the heat transfer limit at this moment. However, after 1400 s the average temperature of the evaporation section of the PHPs with MEPCM suspension as the working fluid is lower than that of the PHP with water. The calculation results show that at this moment the temperature at T1 in the evaporation section with the MEPCM suspension as the working fluid is actually in the phase change temperature range of the MEPCM, causing the temperature at T1 lower than that of the PHP with water, so the average temperature of the evaporation section is lower.

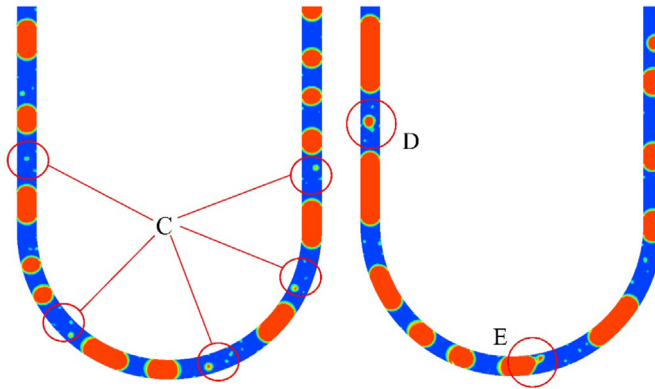
In order to further study the effect of microcapsule particles with different phase change temperature ranges on the anti-dry-out ability of the PHP, the phase change temperature ranges are set to (364–374) K and (404–414) K respectively, and other parameters remain unchanged. The average temperature curves of the evaporation section are obtained (Fig. 8). It can be found that the PHP with MEPCM suspension with phase change temperature range of (364–374) K maintains a steady operation state in 2000 s, so the anti-dry-out ability is remarkably improved. This is mainly because the MEPCM suspension absorbs heat through phase change during the transform from the circulating flow state to the dry-out state of the PHP, which prevents the temperature rising further and acts as a “barrier”, so the temperature of the evaporation section is controlled. However, the MEPCM suspension with phase change temperature range of (404–414) K do not undergo phase change during the state transform, so the temperature raise is not suppressed, and the anti-dry-out ability in 2000 s is not improved compared to that of the PHP with water.

Based on the above conclusions, the effect of MEPCM suspensions with phase change temperature ranges from (354–364) K and (374–384) K on the anti-dry-out ability of PHP is further studied. Other parameters remain unchanged (Fig. 8). It can be found that the MEPCM suspension with phase change temperature range of (354–364) K begins to dry-out later than the water, and the temperature is lower than that of the PHP with water after 1400 s, so the anti-dry-out ability is improved. The MEPCM suspension with phase change temperature range of (374–384) K also begins to dry-out later than the water, but the average temperature is close to that of the PHP with water after 1400 s.

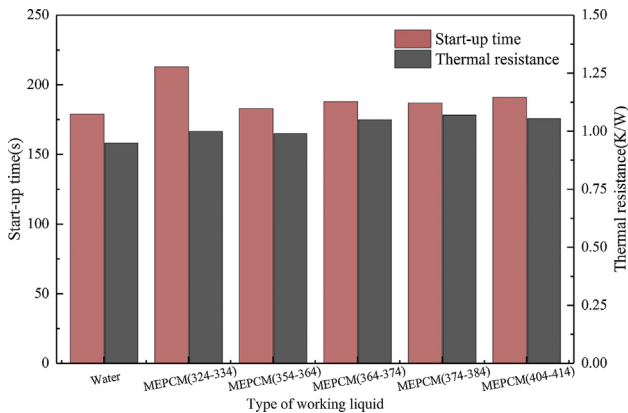
The volume fraction of the vapor phase in each PHP with water and with the above MEPCM suspension as the working fluid at 2000 s is obtained, as shown in Table 2. It can be seen that the PHP with MEPCM suspension with phase change temperature range of (364–374) K has the smallest vapor volume fraction, and the PHP with MEPCM suspension with phase change temperature range of (404–414) K has the largest vapor volume fraction, which is mainly because both the density



(a) Overall volume fraction cloud map of the vapor phase



(b) volume fraction cloud map of the vapor phase of the evaporation section

Fig. 6. volume fraction cloud map of the vapor phase.**Fig. 7.** Start-up time and thermal resistance of the PHPs at 1000 s.

ρ_b and apparent specific heat $C_{p,a}$ of the MEPCM suspension are less than those of the water. When the heating time reaches 3500 s, most of the liquid in the PHP with water is dried out, while the volume fraction of the liquid phase in the PHP with MEPCM suspension with phase change temperature range of (364–374) K still remains 22% (Fig. 9). Therefore, in summary, this MEPCM suspension has the strongest anti-dry-out ability. When MEPCM is used to improve the anti-dry-out ability of PHPs, the phase change temperature of the MEPCM should be selected according to the temperature of the evaporation section when the PHP transforms from the circulating flow state to the dry-out state.

In order to analyze the effect of input powers on the anti-dry-out ability of the PHP, the working process of the PHPs with water and MEPCM suspension with phase change temperature range of (364–374) K at different input powers are calculated and compared in Fig. 10. It can be seen that the higher the input power, the faster the dry-out process of the PHP. At different input powers, the gradient of the

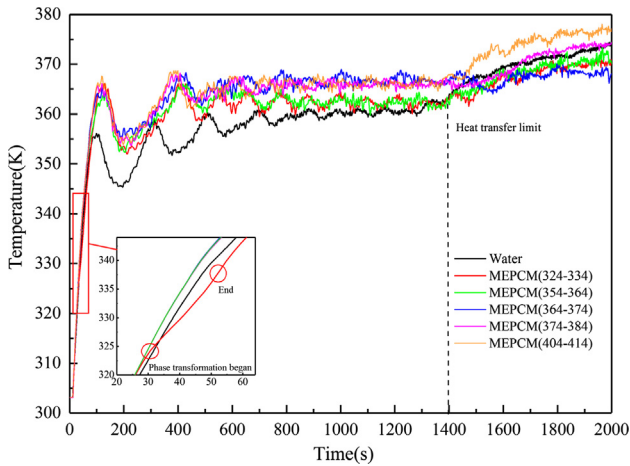


Fig. 8. Temperature changes of the evaporation section of the PHPs.

temperature curve of the PHP with MEPCM suspension is significantly smaller than that of the PHP with water, so the anti-dry-out ability of the PHP with MEPCM suspension is better than that of the PHP with water.

When the equipment needs to be running for a long time, extremely hot conditions may be encountered. If the PHP is dried out and the heat cannot be exported in time, the equipment may be damaged or even result in serious accidents. If the anti-dry-out ability of the PHP under extremely hot working conditions can be improved by adding MEPCM in the PHP, the heat could be exported as much as possible, and thereby the probability of accidents could be reduced.

5. Conclusion

In order to improve the anti-dry-out ability of the PHP, the CFD models of the PHP and the MEPCM suspension are established in this paper, and the effect of the phase change temperature of the MEPCM on the anti-dry-out ability, start-up performance and heat transfer performance of the PHP is studied when the MEPCM suspension is used as the working fluid. Suggestions on the application of MEPCM suspension in PHPs are provided. The conclusions are summarized as follows:

- (1) When the PHP transforms from the circulating flow state to the dry-out state, the MEPCM suspension with phase change temperature range of (364–374) K can delay the temperature raise by heat-absorption in the phase change process, and significantly improve the anti-dry-out ability of the PHP. The appropriate phase change temperature range should be chosen around the average temperature of the evaporation section when the PHP transforms from the circulating flow state to the dry-out state.
- (2) After adding MEPCM in the PHPs, the start-up time of all the PHPs is prolonged. The start-up time of the PHPs with MEPCM suspension as the working fluid is increased by 2.2–18.9% compared with that of the PHP with water. After the MEPCM suspension is used as the working fluid, the thermal resistances of all the PHPs are slightly increased. The observed minimum and maximum thermal resistances are 0.99 K/W and 1.06 K/W, respectively, compared with 0.95 K/W of the PHP with water.
- (3) The numerical method used can well simulate the flow field and phase change process of the vapor phase and the liquid phase in the

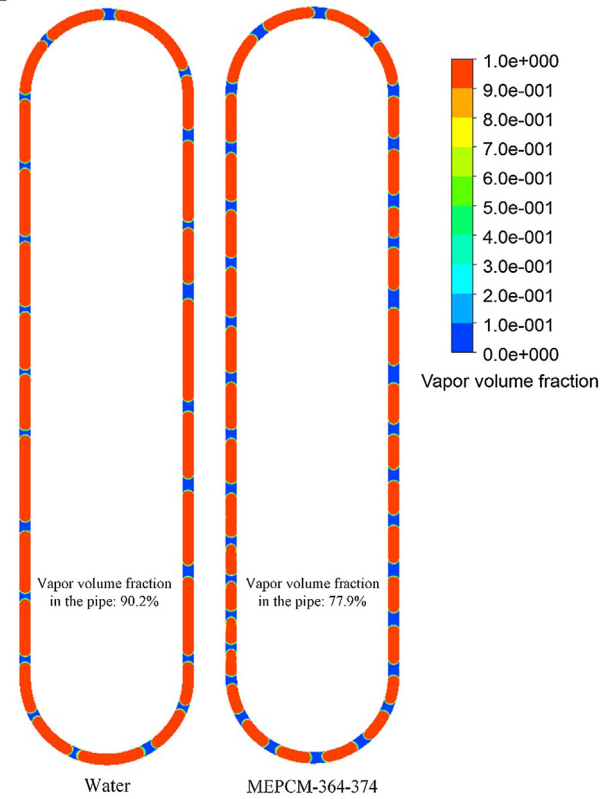


Fig. 9. Overall volume fraction cloud map of the vapor phase with different working fluids at heating time of 3500 s.

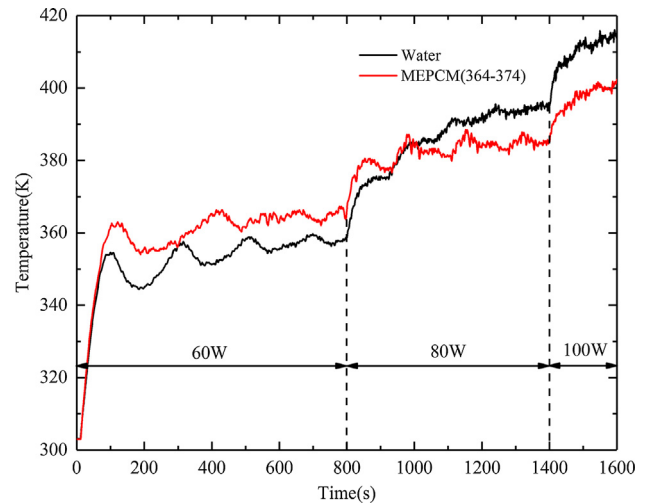


Fig. 10. The temperature changes of the evaporation section at different input powers.

PHP, the endothermic and exothermic effect in the phase change process of the MEPCM, and the variation of the thermodynamic parameters of the MEPCM suspension with the volume fraction of the liquid phase, such as density, specific heat capacity, thermal conductivity and viscosity.

Table 2

Volume fraction of the vapor phase in the PHPs at heating time of 2000 s.

Type of working fluid	water	MEPCM (324–334)	MEPCM (354–364)	MEPCM (364–374)	MEPCM (374–384)	MEPCM (404–414)
Vapor volume fraction	80.1%	76.2%	76.8%	73.1%	77.5%	82.3%

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.applthermaleng.2019.113854>.

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